

The Silver Age of Mercenaries

How Population Decline Drove a Return to Mercenary Armies

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Which Kind of Mercenaries are we Talking About?

Mercenaries as Substitute

Mercenaries as Technology

Why Call It a Silver Age?

- This paper's name pays homage to "The Golden Age of Mercenaries" by Dr. Peter Leeson and Dr. Ennio Piano (2021)
  - Their paper coined the term "Golden Age of Mercenaries" to describe the Mid 13th-Mid 15th century in Europe, in particular Italy
- My paper advances their model of the political-economy behind mercenary demand by rulers to explain the **ERAS** of mass mercenary substitution.
  - Population decline is a second, independent driver, alongside regime popularity.
- The term "Silver Age of Mercenaries," coined by this paper, refers to the resurgence in mercenary employment in Europe from 1618 to 1714
  - By explaining why we see eras of mercenary usage, this paper provides additional modern relevance, as we are seeing an uptick in mercenary employment globally



The Golden Age (Mid 13th- Mid 15th century)



Era of Professionalization (Mid 15th-16th century)

The Silver Age of Mercenaries (1618-1714)

- Despite earlier professionalism, Mercenary usage saw a resurgence across Europe
- For example, the Army of Sweden during the 30 years war consisted between 47% and 82% mercenariness in any given battle



Baseline Model: Military Composition under Political Insecurity

Leeson and Piano (2021)

Setup

- The ruler needs a fixed number of soldiers and chooses the mix: citizens (share c) or mercenaries (share 1 − c).
- Perfect substitutes: the same fighting ability and the same cost.
- Each ruler takes mercenary supply as given.

Notation

w: ruler's wealth while in power

v: severity of mercenary opportunism

g: share of citizens who support the regime

The tradeoff

Mercenaries

- Cannot depose the ruler.
- But they are opportunistic: extortion and desertion cost him wv(1 − c).

Citizens

- No opportunism cost.
- But the share 1 − g who oppose the regime can depose him if armed, with probability c(1 − g).
- A deposed ruler earns nothing.

Optimization and implication

Ruler's problem

$$\max_{c} \left[1 - c + cg\right]\left[w - wv(1 - c)\right]$$

Interior solution

$$c^* = \frac{g(v-1) + 1 - 2v}{2v(g-1)}$$

Comparative static

$$\frac{\partial c^*}{\partial g} > 0$$

- More political security, more citizens.
- Less political security, more mercenaries.

Golden Age model: holding military need and soldier quality constant, lower regime support lowers the optimal citizen share and raises reliance on mercenaries.

Extension: Demography and the Recruitment Base

Remien (2026)

What changes

- Add R, the size of the ruler's domestic recruitment base. R = 1 is the unconstrained baseline; R < 1 is demographic contraction.
- R measures recruitable domestic manpower, not raw population.
- Equal cost is **relaxed**: citizens carry a recruitment-cost premium (in red at right), which is zero at R = 1.
- **Kept:** perfect substitutes, a fixed military need, and mercenary supply taken as given.

Economic mechanism

Holding regime support and military need fixed

- As R falls, domestic soldiers become relatively more expensive compared to mercenaries.
- Same support, same need, different optimal force.

Interpretation

- Baseline: mercenaries rise when rulers distrust citizens.
- Extension: they also rise when citizens become scarce.

Optimization and implication

Ruler's problem

$$\max_{c}[1 - c + cg]\left[w - wv(1 - c) - c\varphi\left(\frac{1}{R} - 1\right)\right]$$

Interior solution

$$c^* = \frac{w[g(v - 1) + (1 - 2v)] + \varphi\left(\frac{1}{R} - 1\right)}{2(g - 1)\left[wv - \varphi\left(\frac{1}{R} - 1\right)\right]}$$

Comparative statics

$$\frac{\partial c^*}{\partial R} > 0 \qquad \frac{\partial c^*}{\partial g} > 0$$

- A smaller recruitment base lowers the optimal citizen share, holding g fixed.
- **Nesting:** at R = 1 the premium vanishes and c* reduces to Leeson and Piano's solution.

Silver Age extension: holding regime support and military need constant, demographic contraction lowers the optimal citizen share and raises reliance on mercenaries.

Venetian Natural Experiment

- Venice fought the Ottomans twice, defending an island fortress each time
  - Fourth War (1570–1573): Cyprus
  - Fifth War (1645–1669): Crete
- **Same enemy.** Same military problem: hold a fortified island in the eastern Mediterranean
- **Same regime:** a stable, long-lasting, high-support Republic, Leeson and Piano's own high-regime-popularity case
- **What changed:** the plague of 1575–77 cut Venice's population from an estimated 170,000 to 134,871 by 1581, about 21%, and the city itself did not exceed its pre-plague population until 1951
- **My extension predicts** that a smaller recruitment base means more mercenaries



Sources: Davis (1962), as reported in Weiner (1970); Clark (2026).

Results: Mercenaries Rose from 38% to 92%

Remien (2026)

A soldier recruited from outside the Venetian realm counts as a mercenary.

Fourth War: Cyprus, 1570–71



Fifth War: Candia, September 1658





Venetian sea fortress, Candia (Heraklion), Crete

Fourth War

5,850 of 15,400 defenders were foreign mercenaries (Hill 1948)

Fifth War

2,200 of 2,400 defenders were foreign mercenaries (Mugnai 2018, p. 118)







Modern Relevance



What's Next?

- I have received feedback from Ennio Piano, one of the authors of the original "Golden Age" paper
- I plan to integrate his comments, along with comments from the scholars here into the manuscript
- Once fully incorporated I will be submitting this manuscript either to *Public Choice* or *European Review of Economic History*